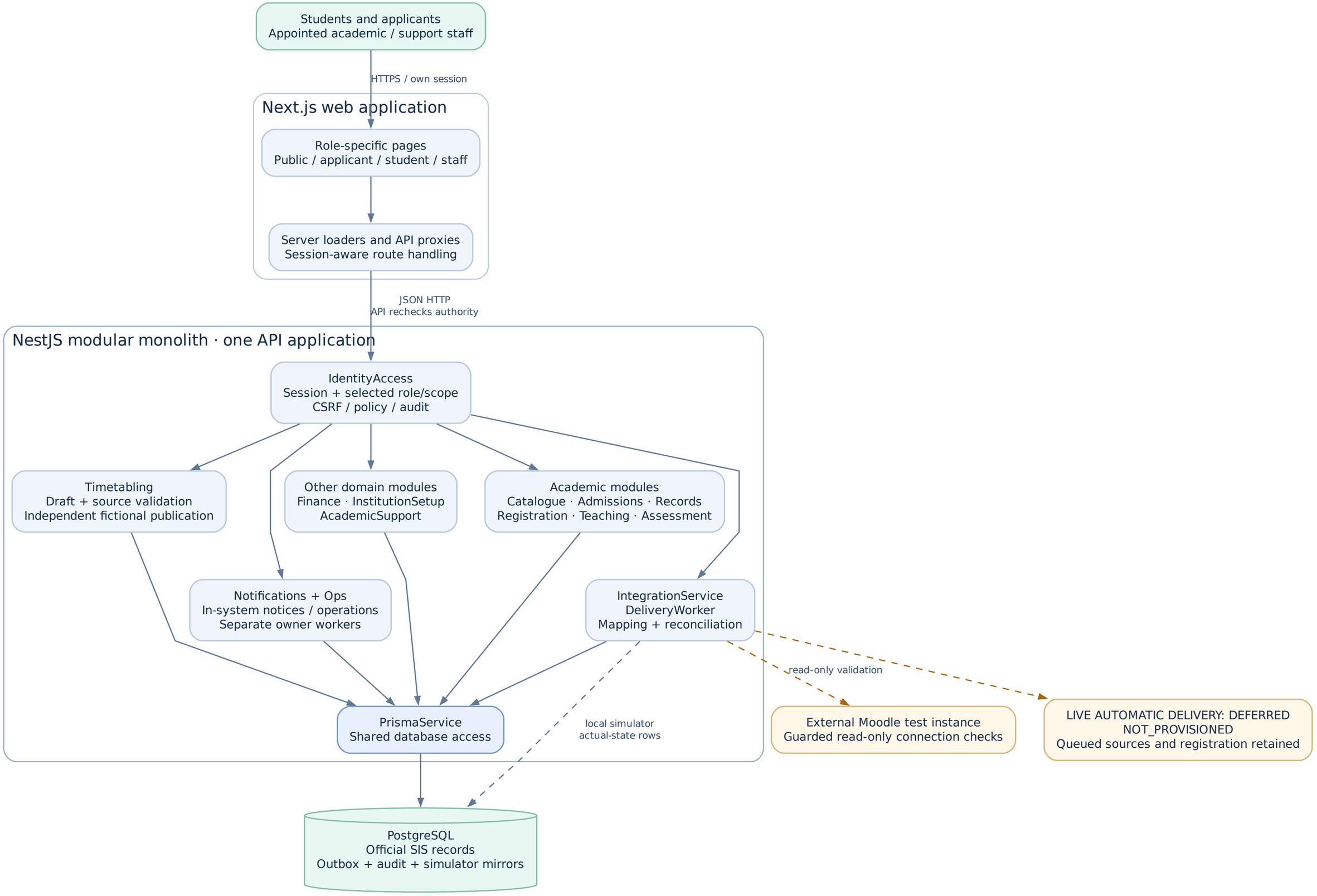
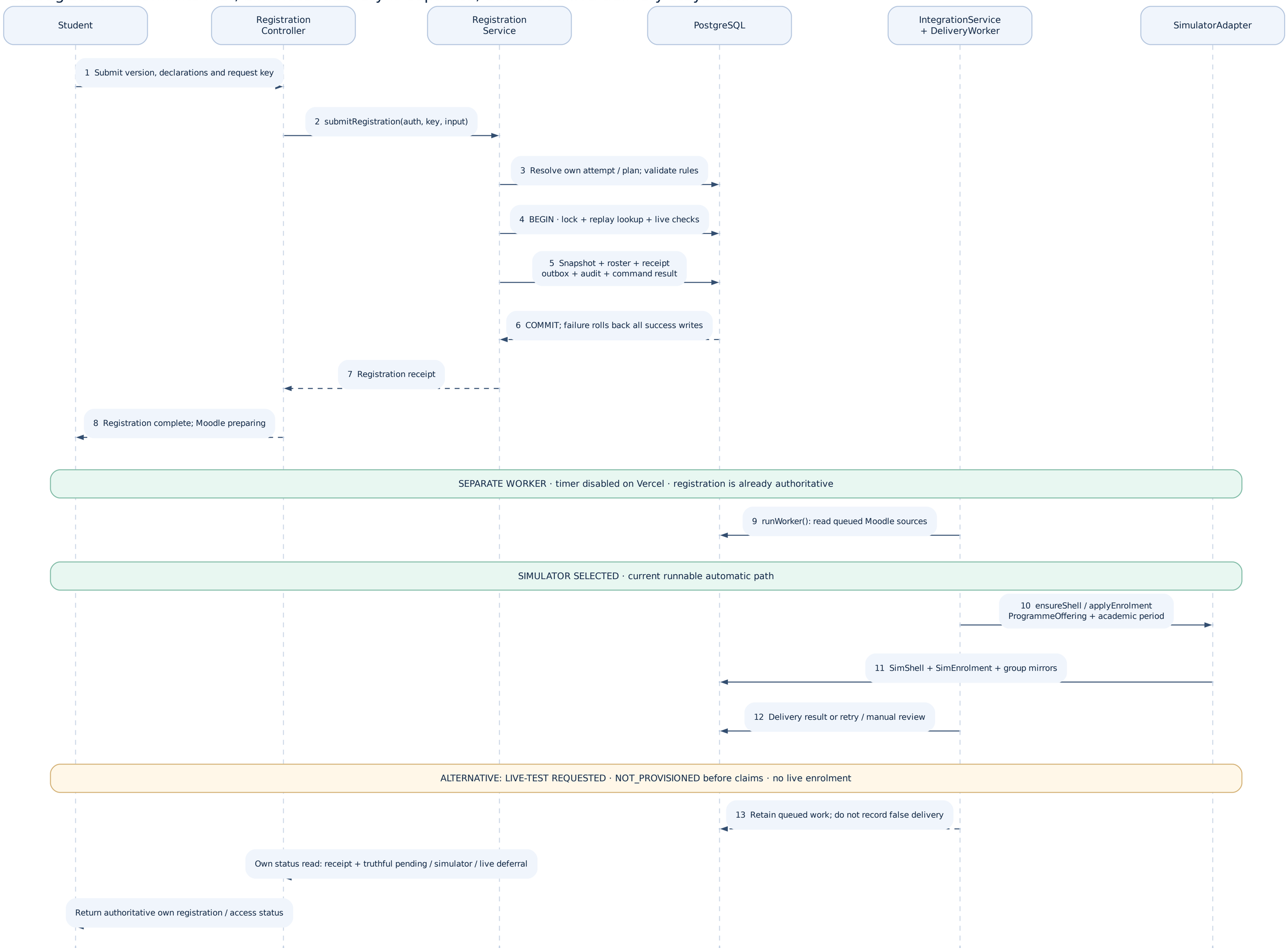
System context and component architecture
Current repository structure · modular monolith · 16 October 2026



Registration and the Moodle handoff

SIS registration commits first; simulator delivery is separate; automatic live delivery stays deferred



Selected domain classes

Solid = declared Prisma relationship · dashed = scalar / polymorphic service reference

Person
id : UUID
displayName; status

1 personId 0..*
1 unique personId 0..1

Account
id; personId
username; status

ProgrammeOffering
id; programmeId
intake; studyMode; campus
Admissions offering

Student
id; personId UNIQUE
studentNumber; status

CurriculumVersion
id; programmeId (scalar)
version; status

Course
id; code; title
credits; capacity

1 grantee 0..*

RoleAssignment
id; accountId
role; scopeType; scopeRef
startsAt; endsAt; revokedAt
capabilities[]

USER: account key no FK

SHELL: offering + period polymorphic, no FK

1 offeringId 0..*
1 studentId 0..*
0..1 curriculum pin 0..*

ProgrammeAttempt
id; studentId; offeringId
curriculumVersionId?
status; applicationId (scalar)

AcademicPeriod
id; code
registrationOpen / Close
status

1 courseId 0..*

CourseVersion
id; courseId; version
title; credits; status

1 courseId 0..*

MoodleMapping
id; kind; sisType; sisId
moodleId; version; status
Polymorphic, not an offering FK

1 scalar attemptId 0..*
1 scalar periodId 0..*
1 periodId 0..*
1 courseVersionId 0..*

InstitutionalRegistration
id; attemptId; periodId
UNIQUE(attemptId, periodId)
version; snapshot; receipt

CourseDeliveryOffering
id; courseVersionId; periodId
deliveryUnitId; campusUnitId
deliveryMode; capacity; status

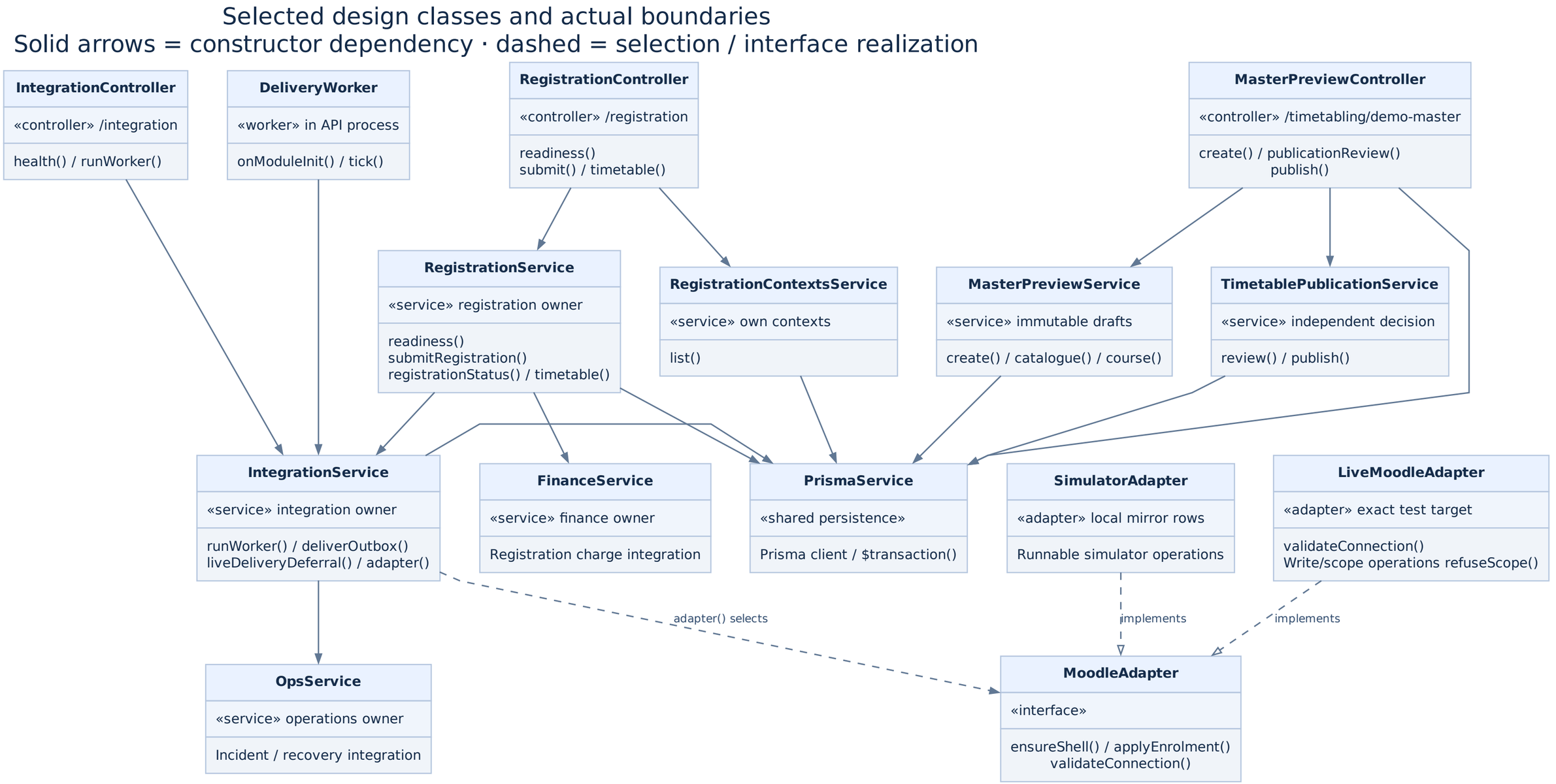
1 own attempt 0..*
1 registrationId 0..*
1 offeringId 0..*

CourseRegistration
id; registrationId; courseId
status = ENROLLED / DROPPED

TeachingSection
id; offeringId; code
capacity; deliveryMode

1 registration lineage 0..*
1 section allocation 0..*

TimetableStudentSession
id; publicationId
courseRegistrationId; attemptId
sectionId; offeringId; periodId
Additional source FKs omitted


Selected design classes and actual boundaries
Solid arrows = constructor dependency · dashed = selection / interface realization
IntegrationController
«controller» /integration
health() / runWorker()
DeliveryWorker
«worker» in API process
onModuleInit() / tick()
RegistrationController
«controller» /registration
readiness()
submit() / timetable()
MasterPreviewController
«controller» /timetabling/demo-master
create() / publicationReview()
publish()
RegistrationService
«service» registration owner
readiness()
submitRegistration()
registrationStatus() / timetable()
RegistrationContextsService
«service» own contexts
list()
MasterPreviewService
«service» immutable drafts
create() / catalogue() / course()
TimetablePublicationService
«service» independent decision
review() / publish()
IntegrationService
«service» integration owner
runWorker() / deliverOutbox()
liveDeliveryDeferral() / adapter()
FinanceService
«service» finance owner
Registration charge integration
PrismaService
«shared persistence»
Prisma client / $transaction()
SimulatorAdapter
«adapter» local mirror rows
Runnable simulator operations
LiveMoodleAdapter
«adapter» exact test target
validateConnection()
Write/scope operations refuseScope()
adapter() selects
implements
implements
OpsService
«service» operations owner
Incident / recovery integration
MoodleAdapter
«interface»
ensureShell() / applyEnrolment()
validateConnection()

Governed fictional timetable publication
Activity and state labels · no in-place promotion of an immutable blocked draft

